



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) EXAMINATIONS - JANUARY/FEBRUARY 2023 SUBJECT : ECE 2155 - SIGNALS AND SYSTEMS (MAKEUP)

SIGNALS AND SYSTEMS [ECE 2155]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Identify whether the following signals are energy signals or power signals. Also, calculate energy and power of the following signals.

a) $x[n] = \left(\frac{1}{3}\right)^n u[n]$

b) $x[n] = \begin{cases} n^2, & 0 \leq n \leq 3 \\ 10 - n, & 4 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases}$ (4)

- 1B) Given $v(t) = \sin(t)u(t)$, identify whether the signal $x(t) = v(t) + v(-t)$ is periodic. Justify your statement (3)

- 1C) A trapezoidal pulse is shown in Figure 1C. Sketch the following signals:

- a) $x(4t)$
b) $x(0.2t)$
c) $x(10t - 4)$

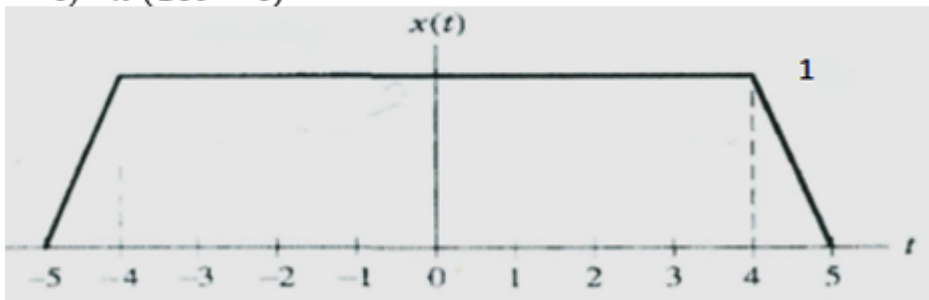


Figure 1C

- 2A) Determine the response of the system described by $y[n] = \frac{1}{3}\{x[n+1] + x[n] + x[n-1]\}$ to the input signal, $x[n] = \begin{cases} 3 & -2 \leq n \leq 2 \\ 0 & \text{otherwise} \end{cases}$ (4)

- 2B) Determine the step response of the LTI system whose impulse response is $h(t) = u(t-1) - 2u(t-2) + u(t-3)$ (3)

- 2C) (3)

Justify whether the following system is linear, time invariant, causal and memoryless:
 $y[n] + 2y[n - 1] = x[n + 1]$.

3A) Let $x_1[n] = \left(\frac{1}{2}\right)^{-n} u[-n]$ and $x_2[n] = \left(\frac{1}{\pi n}\right) \sin(\pi n/3)$, Use convolution property of Fourier representation to determine $x_1[n] * x_2[n]$ (4)

3B) Apply necessary properties to determine the time domain representation of the following frequency domain representation. (3)

$$X(e^{j\Omega}) = \left(\frac{1}{1 - \frac{1}{3}e^{-j\Omega}}\right) \left(\frac{1}{1 + \frac{1}{2}e^{-j\Omega}}\right).$$

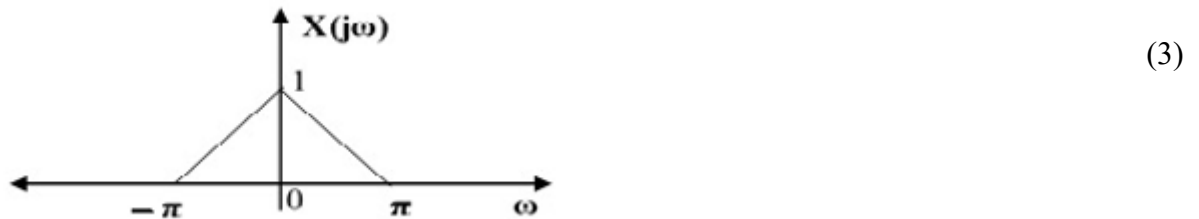
3C) Given $x(t) = \cos(\pi t) + 2 \sin(4\pi t)$. Determine the Fourier representation of $x(t)$ and plot the magnitude and phase spectrum. (3)

4A) Use DTFT to evaluate the response of the discrete time system having impulse response $h[n] = \frac{\sin\left(\frac{\pi}{4}n\right)}{\pi n}$ to the input $x[n] = \cos\left(\frac{\pi}{8}n\right) + \sin\left(\frac{3\pi}{8}n\right)$ (4)

4B) Determine the frequency response and the impulse response of the LTI system which produces output $y(t)$ for the input $x(t)$ as described below: (3)

$$x(t) = e^{-t}u(t), \quad y(t) = e^{-2t}u(t) + e^{-3t}u(t)$$

4C) Continuous time signal $x(t)$ with spectrum as shown in the following diagram is sampled with sampling time $T_s = 0.5$ sec. Zero order hold system is used for the reconstruction. Plot the magnitude spectrum of the reconstructed signal and compare with original spectrum.



5A) Determine the inverse Z-Transform of $H(Z) = \frac{1}{1 - 2.5z^{-1} + z^{-2}}$ for ROC: (4)

(i) $|Z| > 2$ (ii) $|Z| < 0.5$ (iii) $0.5 < |Z| < 2$. For the above specified cases, identify whether the system is causal, stable or both casual and stable.

5B) Determine Z-Transform of the signal and indicate the ROC. (3)

$$x[n] = (0.2)^n [u[n] - u[n - 1]] * n (0.8)^n u[n]$$

5C) For the following system with zero initial conditions described by differential equation, determine the impulse response assuming that the system is (i) stable (ii) causal (3)

$$\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6 y(t) = \frac{d^2}{dt^2} x(t) + 8 \frac{d}{dt} x(t) + 13 x(t)$$

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